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27 July 1979
SUPERSEDING
MIL-V-17848B(SHIPS)
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(See 6.7)

MILITARY SPECIFICATION

VALVES, PRESSURE-REDUCING, STEAM SERVICE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained pressure-reducing valves for steam service.

1.2 Classification. Pressure-reducing valves shall be of the following classes, and compositions as specified (see 6.2.1):

Class A - Internal pilot operated
Class B - Inverted, gas loaded, liquid sealed
Class C - Spring loaded
Composition A - 2-1/4 percent chromium 1 percent molybdenum (maximum temperature 1050°F (see 3.5.6))
Composition B - 1-1/4 percent chromium 1/2 percent molybdenum (maximum temperature 1000°F (see 3.5.6))
Composition D - Carbon steel (maximum temperature 775°F (see 3.5.6)).

Pressure-temperature ratings shall be as specified in 3.5.6, and as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-R-2765 - Rubber Sheet Strip, Extruded, and Molded Shapes, Synthetic, Oil Resistant.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-R-17131 - Rods and Powders, Welding, Surfacing.
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-798 - Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use.
MS16142 - Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions For.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B1.12 - Class 5 Interference-Fit Thread.
B16.5 - Steel Pipe Flanges, Flanged Valves, and Fittings.
B18.2.1 - Square and Hex Bolts and Screws, Including Askew Head Bolts, Hex Cap Screws, and Lag Screws.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A105 - Forgings, Carbon Steel, for Piping Components.
A182 - Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
A193 - Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service.
A194 - Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service.
A216 - Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A217 - Martensitic Stainless Steel and Alloy Steel Castings for Pressure-Containing Parts Suitable for High-Temperature Service.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Class A and B valves furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Sample for first article inspection. For class C valves, prior to beginning production, a sample shall be tested as specified in 4.5 (see 6.4).

3.3 Valve descriptions. Valves shall be of the self-contained, internal operated type. The following is a brief description of the classes A, B, and C:

(a) Class A (internal pilot operated). Downstream pressure feedback is sensed by a spring-loaded diaphragm to position a pilot valve which utilizes the inlet steam pressure to position the main valve plug via an operating piston.
(b) Class B (inverted, gas loaded, liquid sealed). Downstream pressure is controlled by an air (or other inert gas) loaded diaphragm assembly located below the main valve body. Reduced pressure feedback is conducted to the top of the diaphragm, which is protected with a water seal, and compared with the air load to directly position the main valve plug. The bottom surface of the diaphragm is protected with a glycerin seal.

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(c) Class C (spring-loaded). Downstream pressure feedback is sensed by a spring-loaded diaphragm which directly positions the main valve plug.

3.4 Materials. Materials shall be as specified in table I. Materials for parts not listed in table I shall be corrosion-resistant, suitable for the intended pressures and temperatures, and selected to prevent galling, seizing, or excessive wear on operating parts. Clearances shall prevent interferences due to the thermal expansion.

TABLE I. List of materials.

Name of parts	Composition A	Composition B	Composition D
Body, bonnet, and bottom cover[1/]	ASTM A182, grade F22 ASTM A217, grade WC9	ASTM A182, grade F11 ASTM A217, grade WC6	ASTM A105 ASTM A216, grade WCB AISI 1020
Internal trim	See 3.4.1		
Cylinder liner and piston (class A)	400 series corrosion-resistant steel - Parts shall have surface hardness of 500 Brinell minimum		
Gaskets	MIL-G-21032, class B		
Diaphragm (metallic)	Nickel-copper alloy 300 series corrosion-resistant steel Nickel-chrome alloy steel		
Diaphragm (nonmetallic)	MIL-R-2765		
Springs	See 3.4.2		
Bolting[1/]	ASTM A193, grade B16 ASTM A194, grade 4	ASTM A193, grade B16 ASTM A194, grade 2H	ASTM A193, grade B7 ASTM A194, grade 2H

[1/] If desired by the manufacturer, the higher grade bolting materials may be used in lower temperature categories (e.g. ASTM A194, grade 4 may be used for composition B, etc.) and also higher grade body materials for composition B and D valves (e.g. ASTM A182, grade F22 for composition B, etc.).

3.4.1 Trim materials. Unless otherwise specified (see 6.2.1), the valve manufacturer shall select from the categories listed in 3.4.1.1 and 3.4.1.2 (where applicable) the trim materials best suited to meet the requirements.

3.4.1.1 Main valve trim materials. Main valve trim (defined as consisting of the seat or seat ring, and plug and the guide posts and bushings) materials shall be selected from the following:

(a) Stellite - Trim to be Stellite[1/].
(b) Hardened corrosion-resistant steel - Hardened corrosion-resistant steel plug (400 series or 17-4 PH) and Stellite[1/] seat or seat ring. Guiding surfaces to be hardened corrosion-resistant steel or Stellite[1/].

Non-galling grades of materials shall be chosen to prevent galling between rubbing surfaces. A difference in hardness of at least 100 points Brinell shall be maintained between the rubbing surfaces of the guide bushings and posts. This requirements does not apply if both the guide bushings and posts are Stellited or if the hardness of either exceeds 450 Brinell.

[1/] Where Stellite is used, it shall consist of either wrought Stellite 6B, cast Stellite 6, or an inlay of Stellite (not less than 3/32 inch thickness for main seat and disc surfaces). Where inlays are used, welding rods shall be in accordance with type MIL-RCoCr-A of MIL-R-17131.

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3.4.1.2 Pilot valve trim materials (class A). Pilot valve trim (defined as consisting of the seat, valve, and guiding surfaces) shall be made from one or a combination of the following materials:

(a) 400 series or 17-4 PH corrosion-resistant steel - hardened.
(b) Stellite[2/].

3.4.2 Spring materials. Where the working temperature of the spring will exceed 600°F, either Inconel X-750 or A-286 alloy steel shall be used. Where the working temperature of the spring exceeds 450°F, but not 600°F, Inconel 600 or tungsten tool steel may also be used. Where the working temperature of the spring will not exceed 450°F, 300 series corrosion-resistant steel may be used.

3.4.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.5 Design and construction.

3.5.1 Design concept. Valves will be operated, maintained, and repaired on board Naval ships and shall emphasize simplicity, maintainability, ruggedness, and reliability. Design shall permit access for adjustment and repair when working from either side of the valve and without requiring removal of the valve body from the line.

3.5.2 Maintainability. Internal parts shall permit easy disassembly and reassembly with standard tools and shall prevent, as far as practical, the incorrect reassembly of parts. Positioning and alinement of all parts in assembly shall employ positive means so that correct reassembly is repeatedly assured. Parts for a given valve shall not be physically interchangeable or reversible, unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.5.2.1 Valve design shall permit accomplishment of the following maintenance actions within the time limits specified:

Action	Time allowed
Renew pilot valve assembly trim (where applicable)	1/2 hour
Renew main valve trim	3/4 hour

3.5.3 Interchangeability. Valve design shall permit interchangeability without individual modification of like parts between all valves. Each part shall have part number identity and shall be replaceable from stock or the manufacturer on a non-selective and random basis. Parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation (physical) and performance (function). Physically interchangeable assemblies, components, and parts are those which are capable of being readily installed, removed, or replaced without alteration, misalinement, or damage to parts being installed or to adjoining parts. Fabrication operations such as cutting, filing, drilling, reaming, hammering, bending, prying, or forcing shall not be required.

3.5.4 Springs. Springs shall not be fully compressed during any normal operation or adjustment of the valve. The working stress shall be such that relaxation shall not exceed 5 percent over a 1000 hour period at the nominal operating temperature. Spring ends shall be squared and ground.

3.5.5 Threads. Threads shall conform to FED-STD-H28. Where necessary, provisions shall be incorporated to prevent accidental loosening of threaded parts. Pipe threads shall not be used. ANSI B18.2.1 hex-head standards shall be used.

[2/] Where Stellite is used, it shall consist of either wrought Stellite 6B, cast Stellite 6, or an inlay of Stellite (not less than 3/32 inch thickness for main seat and disc surfaces). Where inlays are used, welding rods shall be in accordance with type MIL-RCoCr-A or MIL-R-17131.

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3.5.6 Pressure-temperature ratings. Design and pressure-temperature rating shall be in accordance with ANSI B16.5 except for maximum allowable temperature. Maximum temperature limitations shall be as follows:

(a) Composition A - 1050°F
(b) Composition B - 1000°F
(c) Composition D - 775°F

3.5.7 End preparation. Valves shall be furnished with flanged ends in accordance with ANSI B16.5. Flanges shall be cast or forged integral with the valve body and the inlet and outlet flanges shall be of the same size and pressure rating.

3.5.8 Bonnet and bottom cover joints. Bonnet and bottom cover (where applicable) shall be flanged for attachment to the body. Joints shall be secured by either of the following:

(a) Through-bolts threaded the entire length and fitted with a nut on each end. Threads on both bolts and nuts shall be class 2 fit in accordance with FED-STD-H28.
(b) Tap-end studs with interference fit at the tap end and a class 2 fit at the nut end. Interference fit shall be in accordance with ANSI B1.12.

Bonnet and bottom cover shall be located by body guiding (i.e. a close tolerance fit between machined diameters on the body, bonnet, and bottom cover) rather than depending on studs or bolts for location. Spiral wound gaskets shall be fully retained and the joints shall have metal-to-metal take-up to provide controlled compression of the gaskets. To assure easy gasket removal, not more than two gasket-retaining faces for each gasket shall be formed on a single part. Joint design shall assure parallel alinement of the guide bushings. Sufficient bolting area shall be provided to maintain metal-to-metal make-up over at least a 3 year period. Bearing surface of nuts and their respective surfaces on the valve shall be finished machined.

3.5.9 Body construction. Valve bodies shall be machined from a one-piece casting or forging and shall be of basic globe configuration with in-line inlet and outlet ports. Steam lines, except for the external downstream pressure sensing line (where used), shall be internally ported in the body and bonnet. Body passages shall produce gradual changes in flow direction so as to reduce any effects of concentrated impingement and 90 degree turns. In portions of the valve subject to velocity increases and flow direction changes, such as immediately downstream of the seat, the design shall eliminate direct impingement against the walls at close range.

3.5.10 Control connections. Where external downstream sensing is used, a 1/2 inch iron pipe size (i.p.s.) flanged connection which is either cast or forged integral with the body or bonnet or welded shall be provided.

3.5.11 Internal trim. Internal trim (except welded or brazed in seat rings) shall be readily replaceable without requiring removal of the valve body from the line. The main plug or disc shall be single seated. Guiding of the plug or disc shall prevent binding or seizing and insure proper seating under all design conditions. This requirement shall be maintained with interchangeable parts and under any tolerance stack-up condition. Where quick change cage trim is specified (see 6.2.1), the seat ring shall be gasket sealed and clamped in position by a cage which is either part of or held in position by the bonnet or bottom cover flange. Where cage trim is not specified, and where a separate seat ring is provided, it shall be seal welded or brazed circumferentially on valves used at 225 pounds per square inch (lb/in^2) steam gage pressure and above.

3.5.12 Set point adjustment. Means shall be provided for adjusting the set point through the specified range, with the valve under pressure. The adjusting or loading device shall be safeguarded against accidental change in set point.

3.5.13 Detail class requirements.

3.5.13.1 Class A. The operating piston shall be separate from the main valve and fitted with one or more piston rings. The design shall prevent water build up on the piston. The piston shall operate within a separate hardened-steel cylinder liner located in the valve body so that removal of the valve bonnet provides access to the top of the piston assembly. The cylinder liner shall be held in place by way of the bonnet bolting or shall be permanently fabricated into the body. The requirement to locate the cylinder liner in the body may be waived where it is shown that an alternative location provides a satisfactory maintenance configuration. The pilot valve seat shall be either directly threaded into the bonnet or retained by a gasket-sealed cage construction. Pilot valve and diaphragm chambers

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shall be self-draining. The pilot valve shall be single seated with integral stem. The valve shall be controlled by a spring referenced metal diaphragm and shall open against high-pressure. A return spring shall keep the pilot valve in contact with the diaphragm at all times. The diaphragm shall not travel through center during any phase of operation. Edges contacting the diaphragm shall be rounded to prevent wear and damage. On valves used for steam temperature above 750°F, the main valve return spring shall be recessed in a condensate chamber out of the flow of live steam. The reduced pressure sensing line shall be internal or external as specified (see 6.2.1).

3.5.13.2 Class B. The diaphragm chamber shall be located below the valve body and a water seal shall be provided for the upper surface of the diaphragm and a glycerin seal for the lower surface of the diaphragm to prevent exposure to steam or air. The upper diaphragm chamber shall constitute a reservoir for the supply of water and the design shall not rely on condensation to form an initial protective seal. The construction shall be such that the temperature of the diaphragm does not exceed 180°F with an ambient temperature of 110°F. A loading fitting which permits loading, bleeding, and isolation of the lower diaphragm chamber shall be provided. A pressure gage, with a range from zero to approximately 150 percent of the maximum pressure to which the chamber will be charged to obtain the highest set pressure, shall be provided and attached to the loading fitting. A relief valve shall be provided on the loading fitting to prevent charging the diaphragm chamber beyond its design pressure. Air connections shall be straight thread and O-ring seal construction in accordance with MS16142. Pneumatic attachments shall be protected from external damage or mishandling with a sturdy guard. The entire diaphragm chamber shall be securely attached to the bottom of the valve to prevent any damage from shock or whip. Where specified (see 6.2.1), diaphragm chamber charging equipment, with the necessary tubing and fittings, and suitable for charging to the maximum set point of the regulator, shall be provided.

3.5.13.3 Class C. Valves shall incorporate a metal diaphragm which has sufficient strength and flexibility to meet the specified performance requirements and provide extended service (at least three years) under the operating conditions. Class C valves shall be limited to set gage pressures of 50 lb/in^2 and below.

3.6 Performance requirements.

3.6.1 Accuracy of regulations. The valve shall have an accuracy of regulation, as defined in 6.5.2, as specified (see 6.2.1).

3.6.2 Capacity. The actual steam flow capacity required, in pounds per hour (lb/h), based on the minimum inlet pressure and highest reduced pressure setting under which the valve will be required to operate, shall be as specified (see 6.2.1). The valve shall meet the specified capacity requirement, or any intermediate capacity requirement down to 10 percent of the specified capacity requirement, and shall operate without hunting, chattering, or excessive noise or vibration, or exceeding the accuracy of regulation specified in 3.6.1, under all specified operating conditions.

3.6.3 Range of adjustment. Valves shall be capable of meeting the performance requirements specified in 3.6.1 and 3.6.2 when set at any point within the required range of set pressure adjustment. Unless otherwise specified (see 6.2.1), class A and C valves shall be adjustable over a range of at least 75 to 125 percent of the specified set pressure and class B valves shall be adjustable between 5 and 100 percent of the maximum set point.

3.6.4 Seat tightness. With a dead-end downstream volume not exceeding the volume represented by 100 diameters of downstream pipe, any steam leakage from the inlet to the outlet of the valve shall be limited below a value which will cause a discharge pressure buildup of 10 lb/in^2 in a 1 hour period.

3.6.5 External leakage. There shall be no external leakage which can be detected by use of a mirrored surface and bubble fluid.

3.6.6 Mechanical shock and vibration. Valves shall meet the mechanical shock requirements defined by grade A, class I of MIL-S-901, the HI-shock test guidance of MIL-STD-798, and the environmental vibration requirements defined by type I of MIL-STD-167-1 up to and including 33 Hz.

3.7 Marking.

3.7.1 Body markings. The manufacturer's name or trademark and the body material composition shall be cast or forged integral with the valve body. The size, rating, and a flow arrow shall be cast or forged integral with the valve body or die stamped on raised metal pads (1/8 inch added wall thickness minimum), or stamped on the outside diameter of the flanges.

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3.7.2 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and MIL-P-15024/5. Each valve shall have an identification plate permanently attached to an exposed position on the valve that will not be covered by insulation. The identification plate shall be made of corrosion-resistant material and shall contain the following information or space therefor:

(a) Manufacturer's name and address.
(b) MIL-V-17848 and class.
(c) ANSI rating.
(d) Adjustable set pressure range.
(e) Manufacturer's model and part number or identification.
(f) Manufacturer's drawing numbers (assembly and CD).
(g) Technical manual number.
(h) 9-digit CID/APL number.

3.8 Technical data. The contractor shall prepare engineering drawings and a technical manual in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 3.8.1 through 3.8.2.

3.8.1 Drawings. In addition to the drawing content required by the data ordering documents (see 6.2.2), the following unique features shall be included:

(a) Accurately scaled sectional assembly which clearly depicts the design and construction of the valve.
(b) Bill of material listing specification, grade, condition, and any other data required to fully identify the properties of the materials proposed.
(c) Details of the seat, disc, and stem assembly and all other replaceable internal trim.
(d) Layout of the pressure-containing envelope (body, bonnet, and bottom cap) giving dimensions which control compression of the spiral-wound gaskets. This is to assure that where remachining is necessary to repair the gasket-sealing surfaces on these parts, that compensating cuts can be accurately made to restore original gasket compression. This layout shall also specify the dimensional limits of such corrective remachining within which function of the valve remains unaffected.
(e) Recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies.
(f) Tabulation of required gasket characteristics including all dimensions (with tolerances) and load versus compression characteristics (with tolerances).
(g) Mark areas to be radiographic, magnetic particle, or dye penetrant inspected.

3.8.1.1 Certification data. Certification data sheets shall be prepared by the contractor (see 6.2.2), in addition to the general requirements, the certification data sheets shall include the following:

(a) Application description including (h) and (l) through (o) of 6.2.1.
(b) Valve description.
(c) The set pressure and adjustable range of valve.
(d) Required accuracy of regulation over specified range of operating conditions.
(e) Rated accuracy of regulation over specified range of operating conditions.
(f) Required maximum capacity under specified conditions.
(g) Rated maximum capacity under specified conditions.
(h) Fail-open capacity (for purposes of relief valve sizing).

3.8.1.2 Provisioning data. When specified (see 6.2.1), provisioning data shall be prepared by the contractor (see 6.2.2).

3.8.2 Manuals. In addition to the general requirements for technical manuals (see 6.2.2), the following shall be included as part of the contents:

(a) Drawings for the valve (including certification data sheet). Drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, etc., and may eliminate extraneous lines and details to clarify the interaction of parts.
(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.
(c) Instructions to permit overhaul by shipyard or other repair facility. These should include procedures for checking all critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, etc. Also, the appropriate procedure (that is, part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.

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(d) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.
(e) Adjustment procedure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.2.1).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) First unit examination and tests (see 4.4).
(c) First article inspection (see 4.5).
(d) Quality conformance inspection (see 4.6).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Drawings in accordance with 3.8.1 shall be submitted prior to authorization of qualification tests. Qualification shall consist of the examination and tests specified in 4.3.2 through 4.3.10.

4.3.1 The contractor shall submit drawings (see 3.8.1) prior to qualification tests. Each of the following classes and ratings shall require separate qualification testing. A 1-inch size test valve shall be submitted. Qualification based on the examination and test of that size will then apply to all sizes of that class and rating covered by this specification. Valves of other sizes may be tested. Use of only one valve size for qualification of a class and rating under this specification only applies where the test valve is representative of the basic design features of all sizes of that class and rating for which qualification is desired. The Government reserves the right to determine what are significant variation requiring separate qualification testing.

Class A - 150, 600, and 1500 lb/in^2 (gage) ratings

Class B - 150 and 600 lb/in^2 (gage) ratings

4.3.1.1 Items submitted with test valves. The following items shall be submitted with each test valve.

(a) Internal replaceable trim (throttling elements, poppets, springs, pistons, etc.) - one complete set.
(b) Spiral wound gaskets and piston rings - two complete sets.
(c) Instruction book for installation, operation, and maintenance - one complete set.
(d) One complete set of drawings in accordance with 3.8.1.

4.3.2 Examination. The valve shall be visually and dimensionally examined to determine conformance with the requirements of this specification and the drawings. Particular emphasis shall be placed on the dimensions, finishes, and condition of the guiding and seating surfaces.

4.3.2.1 The maintenance actions specified in 3.5.2.1 shall be demonstrated.

4.3.3 Nondestructive tests. Nondestructive tests shall be as specified in MIL-STD-278 and in accordance with MIL-STD-271. Acceptance criteria shall be in accordance with MIL-STD-278. This shall include RT, MT/PT, pressure, and visual testing.

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4.3.4 Hydrostatic test. Valves shall be tested in accordance with ANSI B16.5. There shall be no external leakage, permanent distortion or structural failure.

4.3.5 Dead-end test. Using steam or air, with an inlet pressure equal to the nominal rating, the outlet pressure in a dead-end volume representing not more than 100 diameters of the downstream pipe, shall not rise more than 10 lb/in^2 in a 1 hour period.

4.3.6 External leakage test. Pressure containing parts, including the diaphragm chamber for class B valves, shall be tested with steam or air to the maximum working pressure to check for external leakage. There shall be no external leakage which can be detected by use of a mirrored surface (for steam) or bubble fluid (for air).

4.3.7 Performance test. The valve shall be subjected to the tests specified in 4.3.7.1 through 4.3.7.3. The delivered pressure shall at all times hold within the limits of plus or minus 5 percent or plus or minus 2 lb/in^2, whichever is greater. There shall be no evidence of hunting, chattering or any other unstable or unsatisfactory operation of the valve during any portion of these tests. The test specified in 4.3.7.1 and 4.3.7.2 shall be conducted with the valve set at the upper and lower limits of the set pressure range (see 3.5.12).

4.3.7.1 Flow-droop test. The flow shall be varied from lock-up to the maximum flow rating of the valve in not more than 15 seconds and from the maximum flow rating to lock-up in not more than 15 seconds. This test shall be conducted under the following sets of conditions:

(a) Maximum inlet pressure - lowest set pressure.
(b) Minimum inlet pressure - lowest set pressure.
(c) Minimum inlet pressure - highest set pressure.
(d) Maximum inlet pressure - highest set pressure.

The maximum inlet pressure for these tests shall be equal to the rating of the valve and the minimum inlet pressure for these tests shall be equal to 75 percent of the rating of the valve.

4.3.7.2 Inlet pressure transient. With a constant flow demand equal to 10, 50, and 100 percent of the maximum, the inlet pressure shall be varied from 75 percent to 100 percent of the inlet rating in not more than 5 seconds and from 100 percent to 75 percent of the inlet rating in not more than 5 seconds.

4.3.7.3 Endurance test. Following the tests of 4.3.7.1 and 4.3.7.2, the valve shall be subjected to a 20 hour operational test to check functioning and performance.

4.3.8 High-impact shock test. The valve shall be subjected to the high-impact mechanical requirements for grade A, class I of MIL-S-901. MIL-STD-798 shall be used for guidance for test procedures.

4.3.9 Vibration test. The valve shall be vibration tested in accordance with type I of MIL-STD-167-1.

4.3.10 Post test examination. After completion of the tests specified in 4.3.3 through 4.3.9, the test valve shall be disassembled and visually and dimensionally examined. Any damage, excessive wear, or signs of galling or pitting shall be cause for rejection.

4.4 First unit examination and tests. The first valve of the same class (class A or B) and rating and intended for the same basic shipboard application, furnished under a contract or order shall undergo the first unit examination and tests specified in 4.4.1 through 4.4.6. First unit examination and tests will be waived on subsequent orders for the same equipment and basic application. First unit tests may also be waived where the manufacturer has sufficient verifiable evidence based on test data or previous shipboard experience or both with the same or similar application to satisfactorily demonstrate that the valve proposed will meet all contract requirements and is suitable for the intended shipboard application. Subsequent valves of the same class and rating and intended for the same basic shipboard application furnished shall undergo quality conformance inspection specified in 4.6.

4.4.1 Examination. Valves shall be examined as specified in 4.3.2.

4.4.2 Hydrostatic test. Valves shall be hydrostatic tested as specified in 4.3.4.

4.4.3 Dead-end test. Valves shall be dead-end tested as specified in 4.3.5.

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4.4.4 External leakage. Valves shall be tested for external leakage as specified in 4.3.6.

4.4.5 Performance test. The tests specified in 4.4.5.1 and 4.4.5.2 shall be conducted with the valve set at the upper and lower setting of the adjustable set pressure range required by the application (see 6.2.1). The maximum inlet temperature, the range of operating inlet pressures, the maximum rate of change of the inlet pressure, the maximum flow capacity required, and the maximum rate of change of flow demand shall be as specified (see 6.2.1) to meet the application requirements. The required accuracy of regulation shall be maintained and there shall be no evidence of hunting, chattering, or any other unstable or unsatisfactory operation of the valve over any portion of the required operational range of the valve.

4.4.5.1 Flow-droop test. The flow shall be varied from lock-up to the maximum flow rating and back at the rate specified (see 6.2.1). This test shall be conducted under the following sets of conditions:

(a) Maximum inlet pressure - lowest set pressure.
(b) Minimum inlet pressure - lowest set pressure.
(c) Minimum inlet pressure - highest set pressure.
(d) Maximum inlet pressure - highest set pressure.

4.4.5.2 Inlet pressure transient. With a constant flow demand equal to 10, 50, and 100 percent of the maximum, the inlet pressure shall be varied over the specified range at the maximum rate specified (see 6.2.1).

4.4.6 Nondestructive tests. Nondestructive tests shall be conducted as specified in 4.3.3.

4.5 First article inspection. First article inspection for class C valves shall consist of the examination specified in 4.3.2 and 4.5.3 and tests specified in 4.4.1 through 4.4.6 and 4.5.1 and 4.5.2. Each test valve shall be submitted with the items specified in 4.3.1.1.

4.5.1 Endurance test. Following the tests specified in 4.4.5, class C valves shall be subjected to a 5-hour operational test to check functioning and performance.

4.5.2 High-impact shock and vibration tests. High-impact shock and vibration tests shall be as specified in 4.3.8 and 4.3.9.

4.5.3 Post test examination. After the completion of the tests specified in 4.5, the test valve shall be disassembled and visually examined. Any damage, excessive wear, or signs of galling or pitting shall be cause for rejection.

4.6 Quality conformance inspection. Each production valve shall be examined as specified in 4.6.1 and tested as specified in 4.3.3 through 4.3.6.

4.6.1 Visual and dimensional examination. Each valve shall be visually and dimensionally examined to verify conformance with the requirements of this specification not requiring tests.

4.7 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging, packing, and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B, or C and marked in accordance with MIL-V-3, as specified (see 6.2.1).

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5.1.1 Repair parts. Repair parts shall be preserved-packaged, packed, and marked in accordance with the levels of MIL-R-196 as applicable for the intended use and destination as follows:

	Levels	
	Preservation-packaging	Packing
On board (accompanying valves)	A	C
Stock	A	B
Immediate use	C	C

Unless otherwise specified (see 6.2.1), repair parts shall be packed separately and shipped concurrently with the valve(s).

5.2 Use of loose-fill material.

5.2.1 For level C packaging and packing. Unless otherwise specified by the contracting activity (see 6.2.1), use of loose-fill material for level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with loose-fill material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.2.2 For level A packaging and level A and B packing. Use of loose-fill material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler, and dunnage.

5.3 Technical manuals. Technical manuals, which accompany shipments that are packed level A or B, shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed within any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall also state the approximate location therein. For ease of removability the location of the manual shall be such that it is readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B, or C, as specified (see 6.2.1).

5.4 Data (see 3.8). Data shall be prepared for delivery in such a manner as to insure the required information is protected against deterioration, physical damage or loss during shipment from the contractor to the receiving activity. Packages or shipping containers shall as a minimum conform to the level C requirements of 5.1.

6. NOTES

6.1 Intended use. Pressure-reducing valves covered by this specification are intended for line steam pressure reducing service on board ship.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, composition, and pressure rating required (see 1.2).
(c) Trim materials where specific requirement is known (see 3.4.1).
(d) Quick change cage trim, when required (see 3.5.11).
(e) Whether internal or external reduced pressure sensing line is required (see 3.5.13.1).
(f) Charging equipment to be supplied with class B valves (see 3.5.13.2).
(g) Accuracy of regulation required (see 3.6.1).
(h) Minimum and maximum inlet gage pressures (lb/in^2) (see 3.6.2 and 4.4.5).
(i) Range of adjustment for valves (see 3.6.3).
(j) If provisioning data is required (see 3.8.1.2).

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(k) Quality assurance requirements (see 4.1.1).
(l) Maximum rate of inlet pressure variation (lb/in^2 per second) where known (see 4.4.5 and 4.4.5.2).
(m) Maximum inlet steam temperature (°F) (see 4.4.5).
(n) Maximum rate of flow demand variation (pounds per hour per second) where known (see 4.4.5).
(o) Maximum and minimum capacity required lb/h (see 4.4.5.1).
(p) Level of preservation, packaging, and level of packing and marking required (see 5.1 and 5.3).
(q) When "loose-fill" is permitted (see 5.2.1).

6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.8.1 and 3.8.1.1	Drawings, engineering and associated lists	DI-E-7031	Level 2 Design activity designation - contractor Drawing number - contractor Delivery of hard copies - contracting activity
3.8.1.2	Supplementary provisioning technical documentation (SPTD)	DI-V-7000	------
3.8.2	Manual, technical, preliminary	DI-M-2043	Type I of MIL-M-15071

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Where the Government has limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings may be marked with a legend. If used, the "Limited Rights Legend" of DAR should be used.

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL 17848 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 7111, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

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6.4 First article inspection. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions. The definitions specified in 6.5.1 through 6.5.7 shall apply to the various technical terms wherever such terms appear in this specification.

6.5.1 Set pressure. The downstream pressure which the valve is set to maintain under a given set of operating conditions (i.e. inlet pressure and flow). Normally the valve is set with a flow of approximately 10 percent of rated capacity.

6.5.2 Accuracy of regulation. The maximum permissible band over which the downstream pressure may vary when the valve is set at any pressure within the required range of adjustment and is subjected to any combination of inlet pressure, flow demand, and ambient temperature variations, within the specified limits.

6.5.3 Set pressure limits. The range of set pressures over which the valve can be adjusted while meeting the performance requirements specified.

6.5.4 Lock-up pressure. The outlet pressure delivered by a reducing valve under shut off conditions (i.e. when the flow is reduced to a point where it is equal to or less than the allowable leakage).

6.5.5 Nominal pressure. The approximate maximum pressure which the valve will be subjected to in service under normal conditions.

6.5.6 Design pressure and temperature. The maximum pressure and temperature the valve should be subjected to under any condition. The pressure and temperature upon which the strength of the pressure containing envelope is based.

6.5.7 Proof pressure. The maximum test pressure that the valve is required to withstand without permanent damage. Valve operation is not required during application of the proof pressure, but after the pressure has been removed, the valve shall be capable of meeting all performance requirements.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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